

1. Three cards are drawn in succession from a deck without replacement. Find the probability distribution for the number of spades.

2. Determine the value c so that each of the following functions can serve as a probability distribution of the discrete random variable X :

$$f(x) = c \binom{2}{x} \binom{3}{3-x}, \text{ for } x = 0, 1, 2.$$

3. The probability distribution of X , the number of imperfections per 10 meters of synthetic fabric in continuous rolls of uniform width, is given by

x	0	1	2	3	4
$f(x)$	0.41	0.37	0.16	0.05	0.01

Construct a cumulative distribution of X .

4. Suppose that the number of cars X that passes through the car wash between 4:00 PM and 5:00 PM on any sunny Friday has the following probability distribution.

x	4	5	6	7	8	9
$p(x)$	1/12	1/12	1/4	1/4	1/6	1/6

Let $g(X) = 2X - 1$ represent the amount of money (in dollars), paid to the attendant by the manager. Find the attendant's **expected** earnings for this particular time period.

5. Calculate the variance of $g(X) = 2X + 3$, where X is the discrete random variable with the probability distribution

x	0	1	2	3
$f(x)$	1/4	1/8	1/2	1/8

6. Solve the following problem.

The probability distribution of a RV X is given below:

x :	0	1	2	3
$p(x)$:	0.1	0.3	0.5	0.1

If $Y = X^2 + 2X$, find the probability distribution, mean and variance of Y .